10/10/2026

Ethical Hacking Course 1

Rubric 1

Introduction:

A hypervisor is the foundational technology that will allow to run multiple independent operating systems on a single physical machine, making it a necessary foundation for homelab or enterprise testing environments. A has multiple advantages, it allows:

- Kali Linux to run inside Windows like a regular app.
- Switch between Windows and Kali without rebooting.
- VirtualBox to use only a portion of your system's hardware for Kali.
- Kali to run smoothly without affecting your main operating system.

There are two types of hypervisor, type 1 runs directly on the physical hardware. Type 2, runs Inside the existing desktop OS. for our lab, we will need hypervisor type 2 to run it on windows laptop. There are many popular examples of hypervisor type 2: VirtualBox, VMware Workstation, and UTM. In my home lab, the hypervisor of choice is the Oracle VirtualBox.

1- Install a VirtualBox

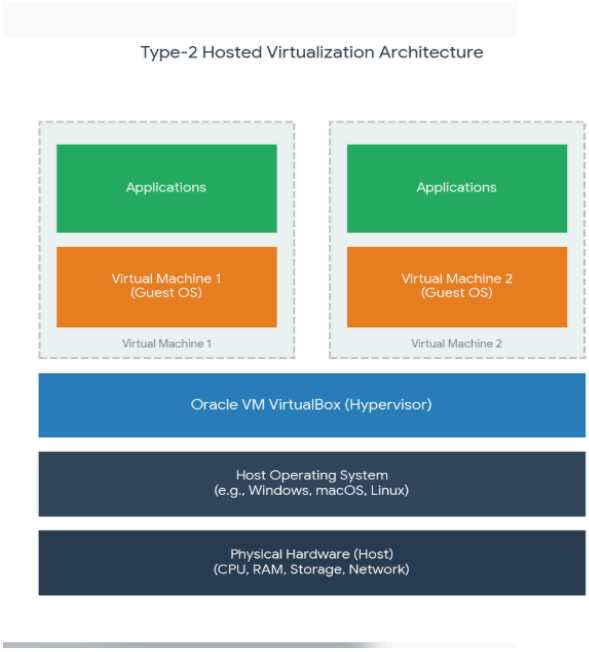
About Oracle virtual box:

Oracle virtual box is a cross-platform virtualization application. It extends the capabilities of the existing computer by running multiple operating systems inside multiple virtual machines. So I can run Linux on my Windows laptop at the same time. Choosing the number of virtual machines to install depends on the practical limits of the disk space and memory.

Note: The rubric requires the installation and documentation of Parrot Virtual machine, but in my lab I have downloaded and installed the Kali Linux virtual machine instead of Parrot (approved substitution by the instructor).

Rubric Completion Requirements

- ✅ Kali + VirtualBox documented
- ✅ Nmap — covered across Weeks 4/6
- ✅ Wireshark — covered across multiple weeks
- 🔶 Metasploit — functional, but strongest evidence (a real session) i
- 🔶 Isolation screenshots — not yet captured
- 🔶 Board task statuses — last known gap was Week 3's checkbox and Week 4/6 open items; worth a quick check



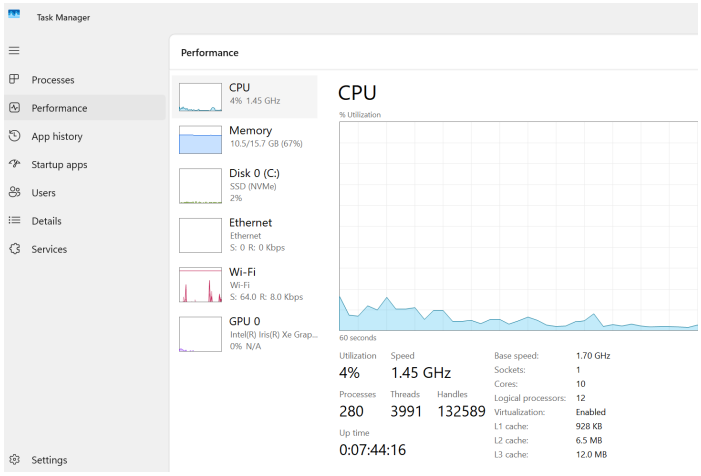
Performing the Installation

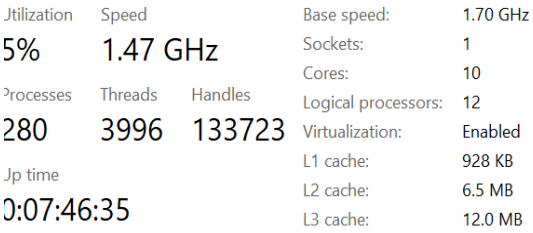
Before starting the download of the Oracle virtuabox, we need to have virtualization enabled in windows on our laptop or PC that we are using. To check it we go to Task Manager by right clicking on the taskbar, then choosing performance and CPU. right on the bottom, it should say virtualization enabled. If it is disabled then, it has to be enabled from the Bios. and some pre-requisites:

64-bit CPU (with 4+ cores)

Disk space (50 GB of free space is needed, but more is preferred)

System memory of a minimum of 8 GB RAM is recommended.



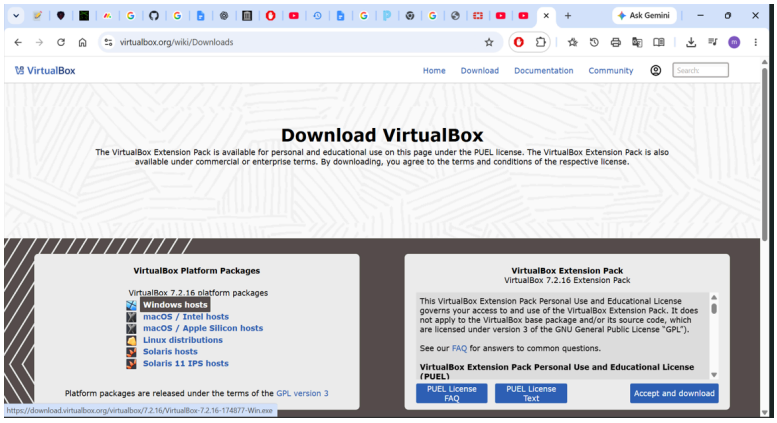


Oracle VirtualBox is available in a number of package formats native to various common Linux distributions. Installing Oracle VirtualBox can be either from a Debian or Ubuntu Package.

Oracle VirtualBox can be downloaded by just typing Oracle Virtual Box on the browser search bar and click on the link

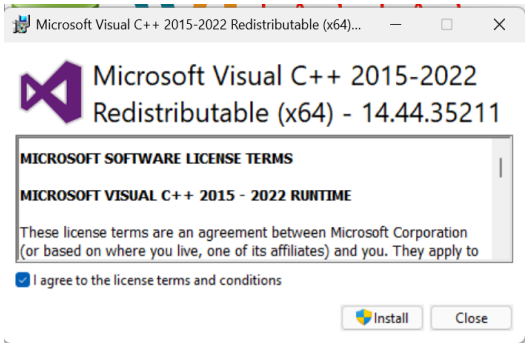
Go to the following URL: https://www.virtualbox.org/wiki/Downloads

Click on it and you will see a “Get Started” section with a download button. We click on the download. It will take us to another page. We need to download the VirtualBox Platform Package and theVirtualBox Extension Pack. In my laptop, I have windows running, so I have to choose windows hosts platform package. If you have a different operating system, choose the appropriate hosts package. And to download the VirtualBox extension pack, on the left side we need to click on accept and download. Once we do that both, the windows hosts package and the extension pack will start downloading.



As of the writing of this module, the latest version of VirtualBox is 7.2.20

This version requires Microsoft Visual C++ 2015-2022 Redistributable.



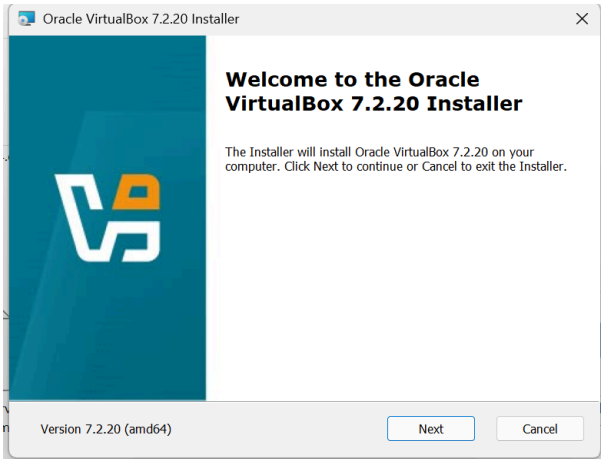
VirtualBox-7.2.20-175154-Win.exe

Oracle_VirtualBox_Extension_Pack-7.2.20.vbox-extpack

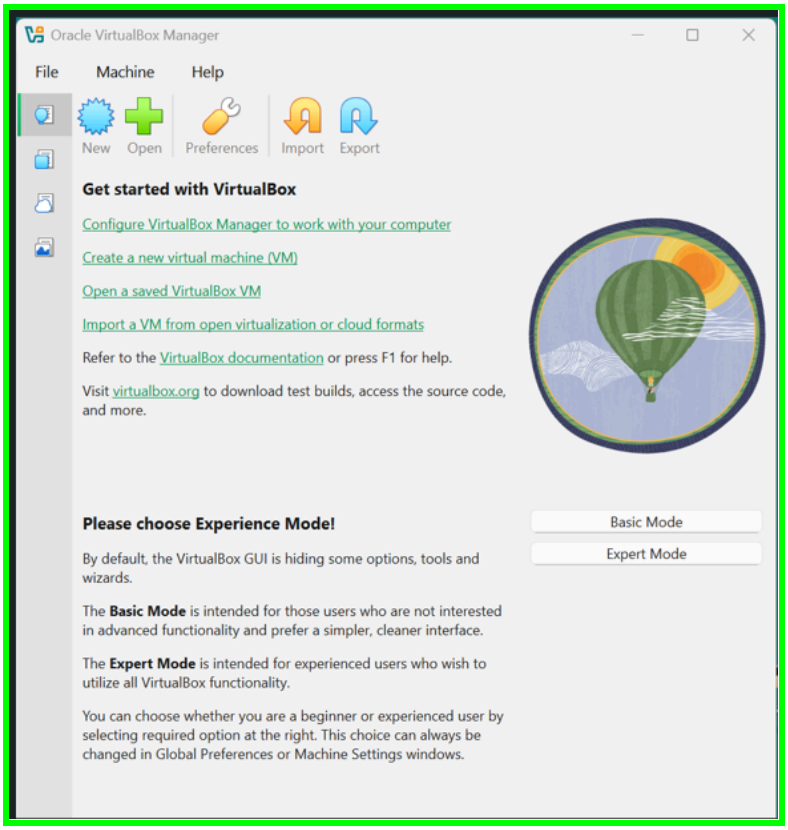
And once the download is complete, we will see these two folders in the download folder.



Now let's go ahead and install Oracle VirtualBox. We just click on virtual Box folder and follow the default install.



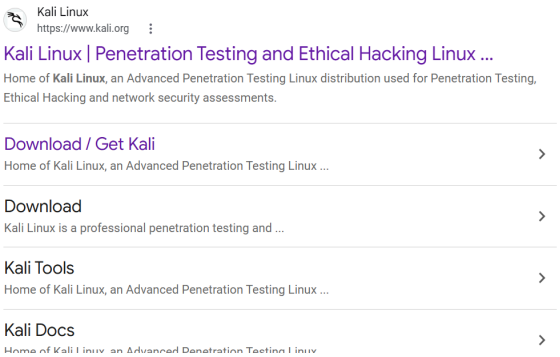
Once the Virtualbox is installed, we go ahead and start it. When we have the Virtual Box manager installed, we can go ahead and close it.



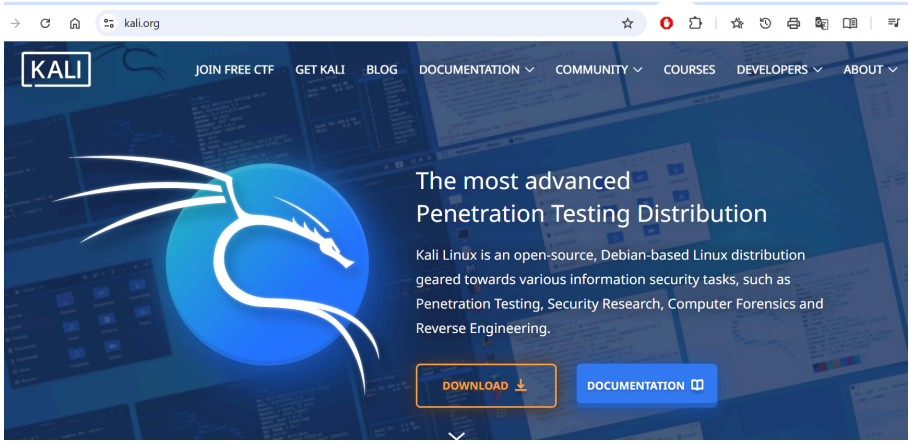
Now we need to go to the VirtualBox extension and right click on it and try to open it with the virtualBox manager. We will then see that the VirtualBox extension pack is installed.

2-Create the Kali Virtual Machine

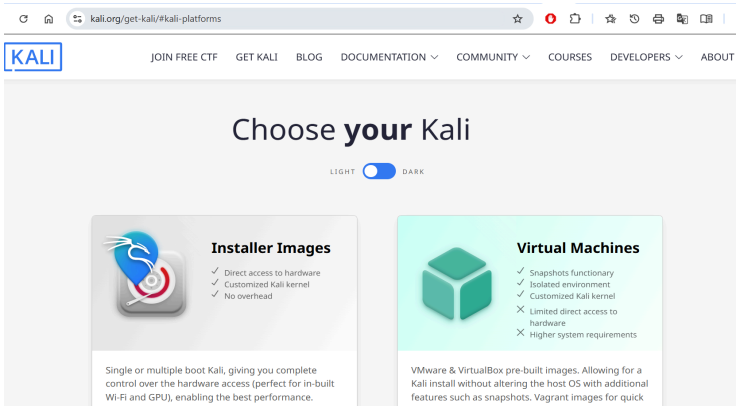
To download Kali Linux Virtual machine, we need to go to https://www.kali.org

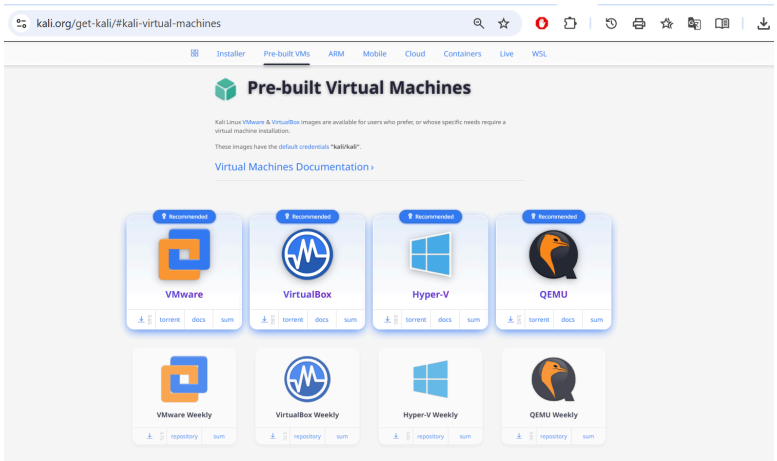


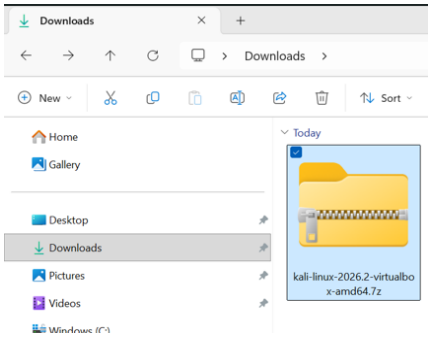
Click on the link, and it will take you to:



By clicking on download, it will take us to where we need to choose the image installer. I will go for the pre-built virtual machine option. We will get to it by scrolling down till I see Virtual Machine. The one in the middle, then click on the download icon. Of course someone else may choose to go with Installer Complete offline with customization option.

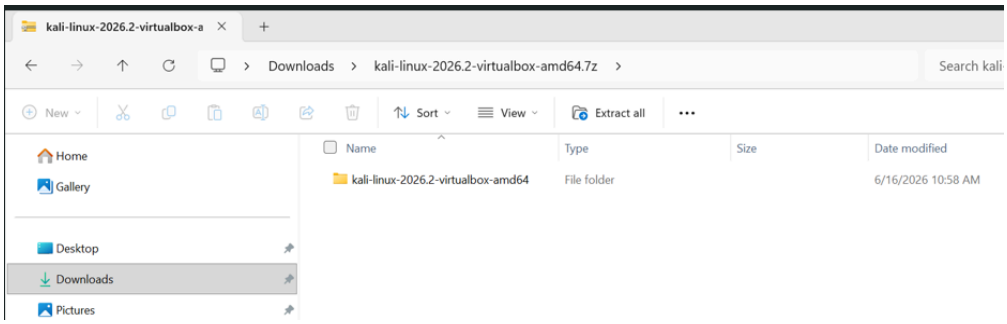






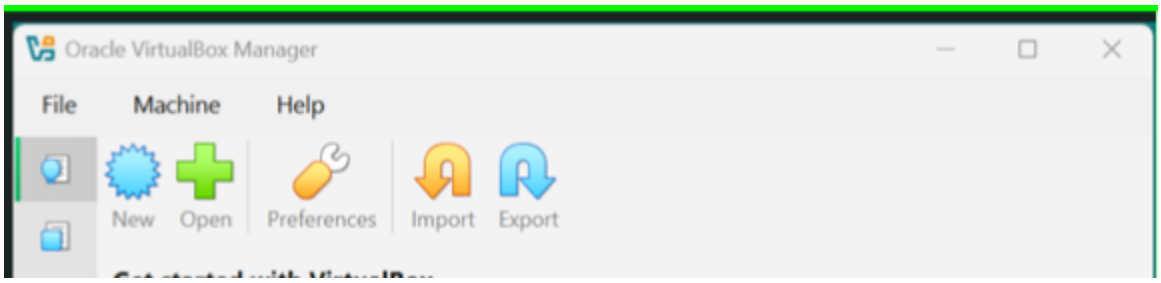
The folder downloaded is a 7z . I needed to unzip the files. Right click on the folder and choose extract all.

This folder shows the unzipped files.



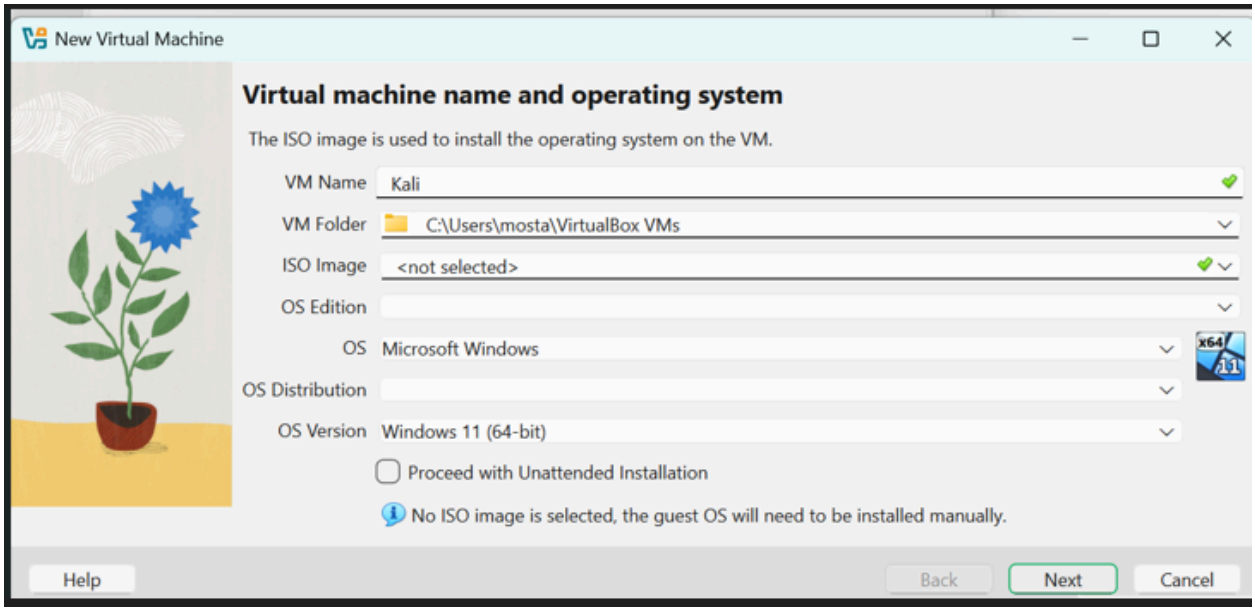
1-Create the Kali Virtual Machine

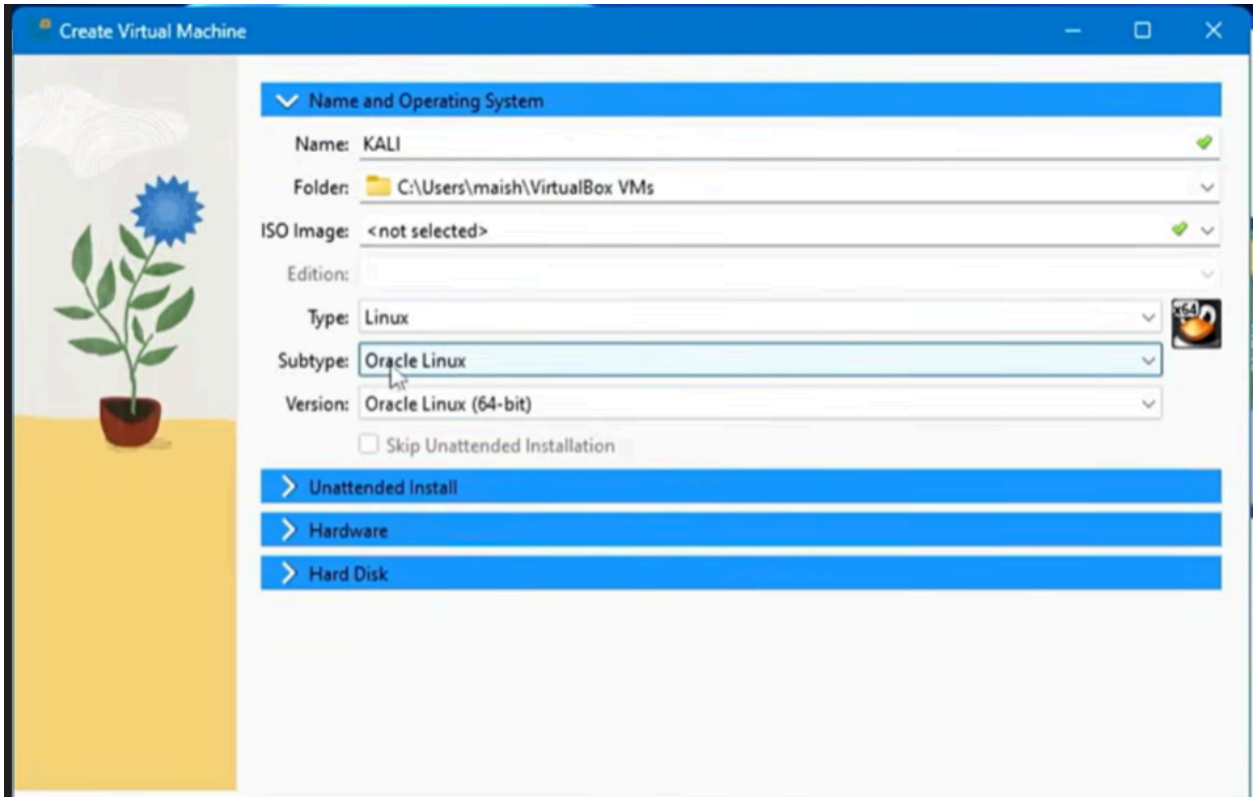
Open the VirtualBox and then press the new button



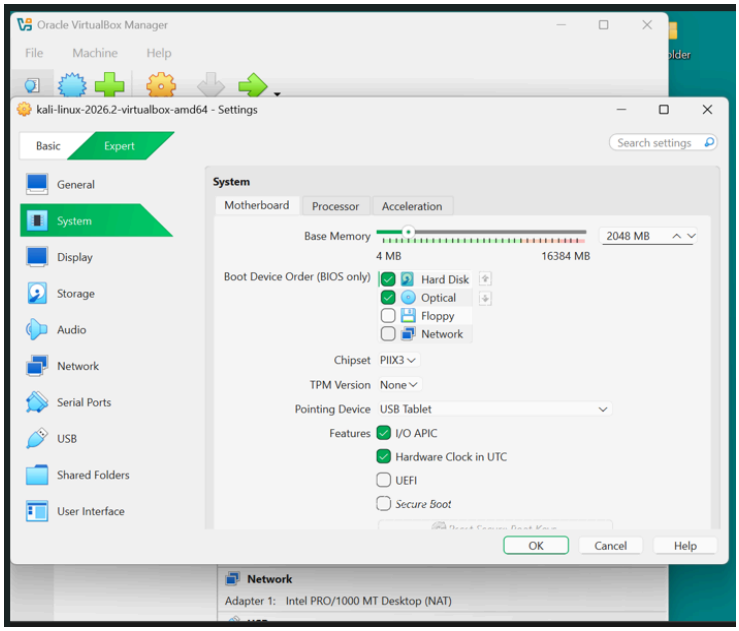
We need to name our virtual machine and select the type

- Type the name of the virtual machine and select it to Linux, then Oracle Linux for Subtype, then in version select: Oracle Linux 64bit.

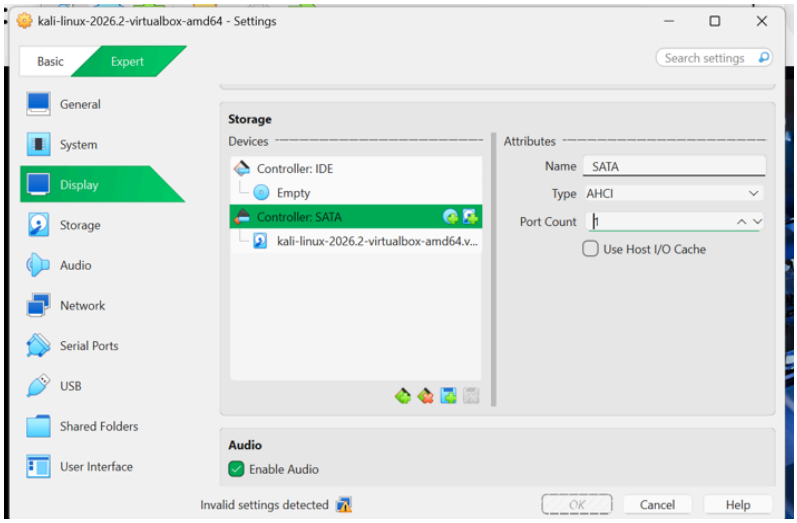




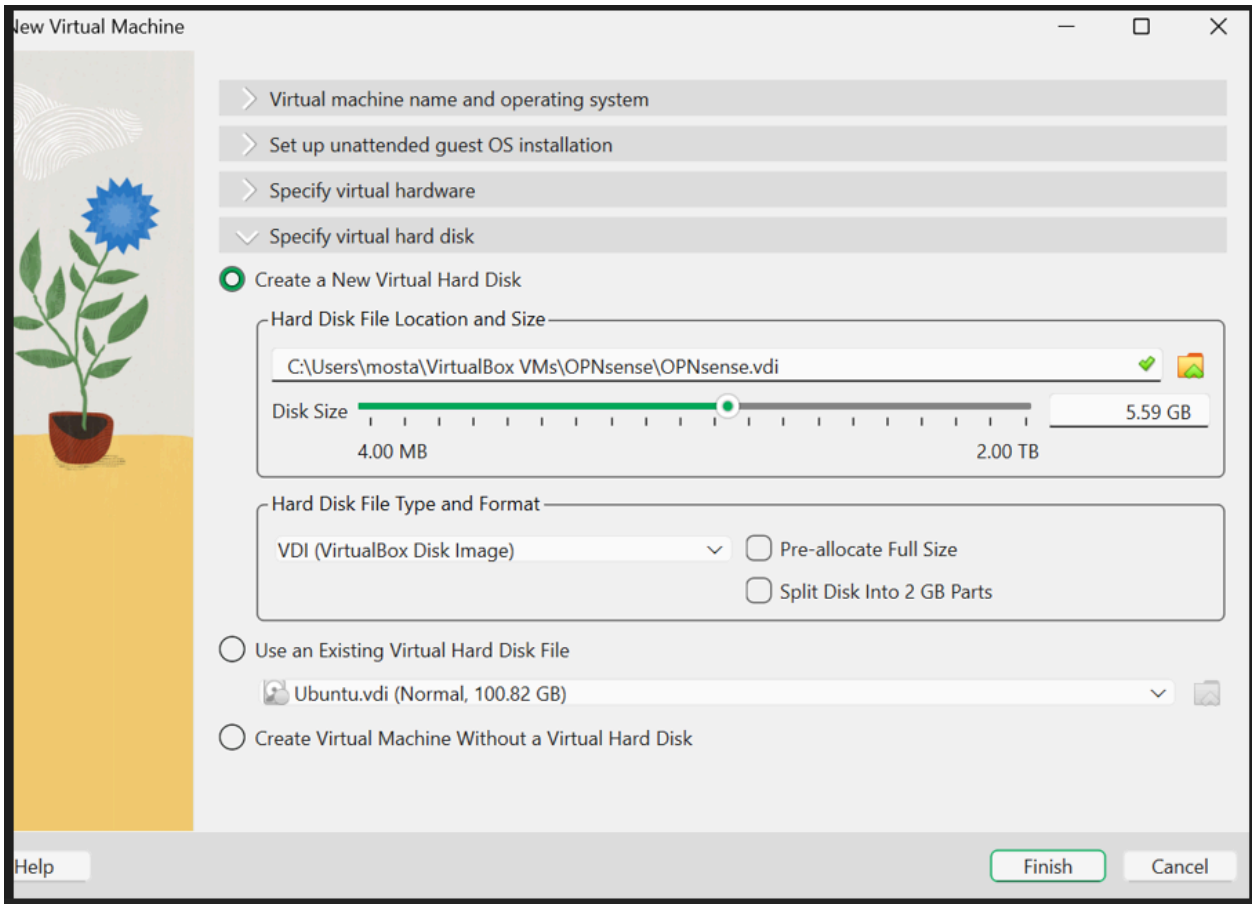
- Allot the size of RAM memory we want to allocate to the virtual machine of Kali Linux. (20000 MB is recommended for normal usage but you allocate more)

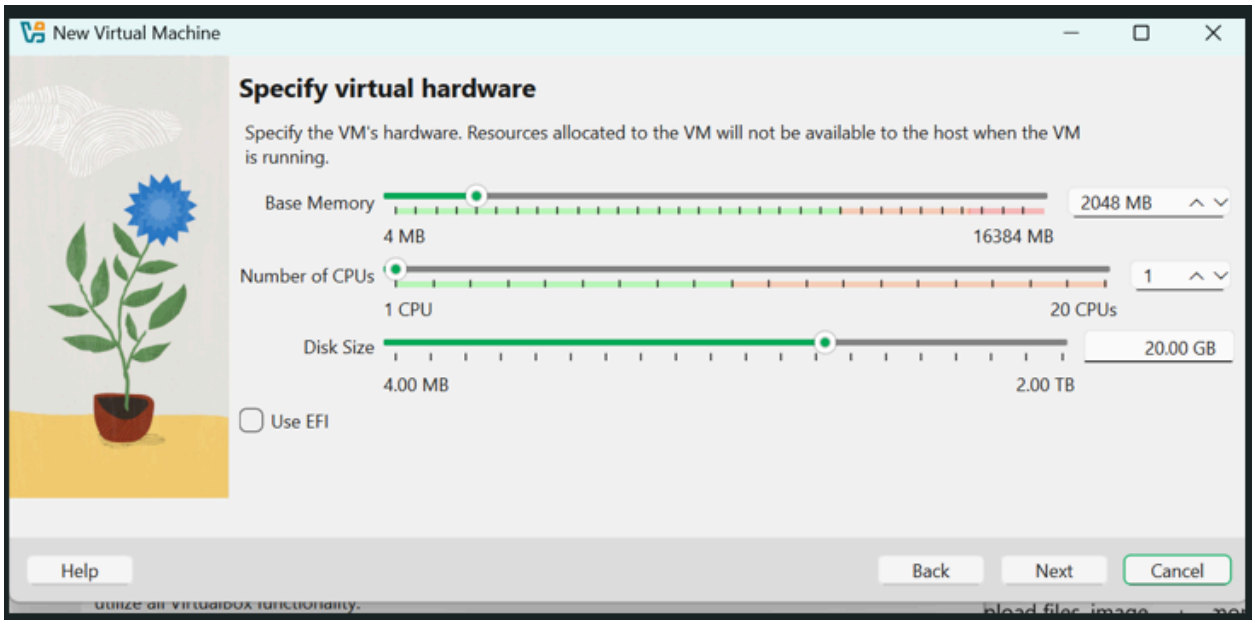


- Select the option to create a virtual hard disk now and then click on the create button.
- Now, Select the Hard Disk FileWrite type for Virtual Hard Disk. (VDI is recommended for daily purposes)

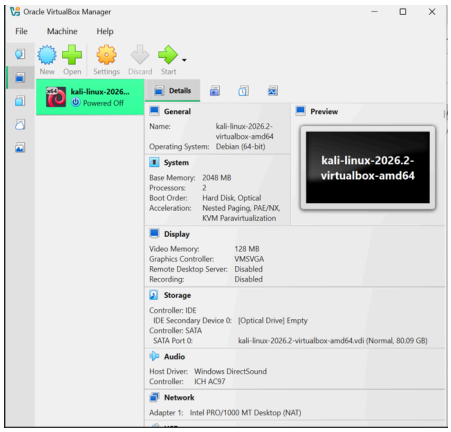


- Select the size of your virtual hard disk and also the location where you want to save your machine and its files.


New Virtual Machine
Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Create a New Virtual Hard Disk
Hard Disk File Location and Size
C:\Users\mosta\VirtualBox VMs\OPNsense\OPNsense.vdi
Disk Size
5.59 GB
4.00 MB
2.00 TB
Hard Disk File Type and Format
VDI (VirtualBox Disk Image)
Pre-allocate Full Size
Split Disk Into 2 GB Parts
Use an Existing Virtual Hard Disk File
Ubuntu.vdi (Normal, 100.82 GB)
Create Virtual Machine Without a Virtual Hard Disk
Help
Finish
Cancel


New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
2048 MB
4 MB
16384 MB
Number of CPUs
1
1 CPU
20 CPUs
Disk Size
20.00 GB
4.00 MB
2.00 TB
Use EFI
Help
Back
Next
Cancel

- Step 2: Start VM and Load ISO
- As soon as the processing is completed click on the Virtual machine name on the left panel and click on the start button from the top.



Configuring Kali Virtual Machine network and ip address.

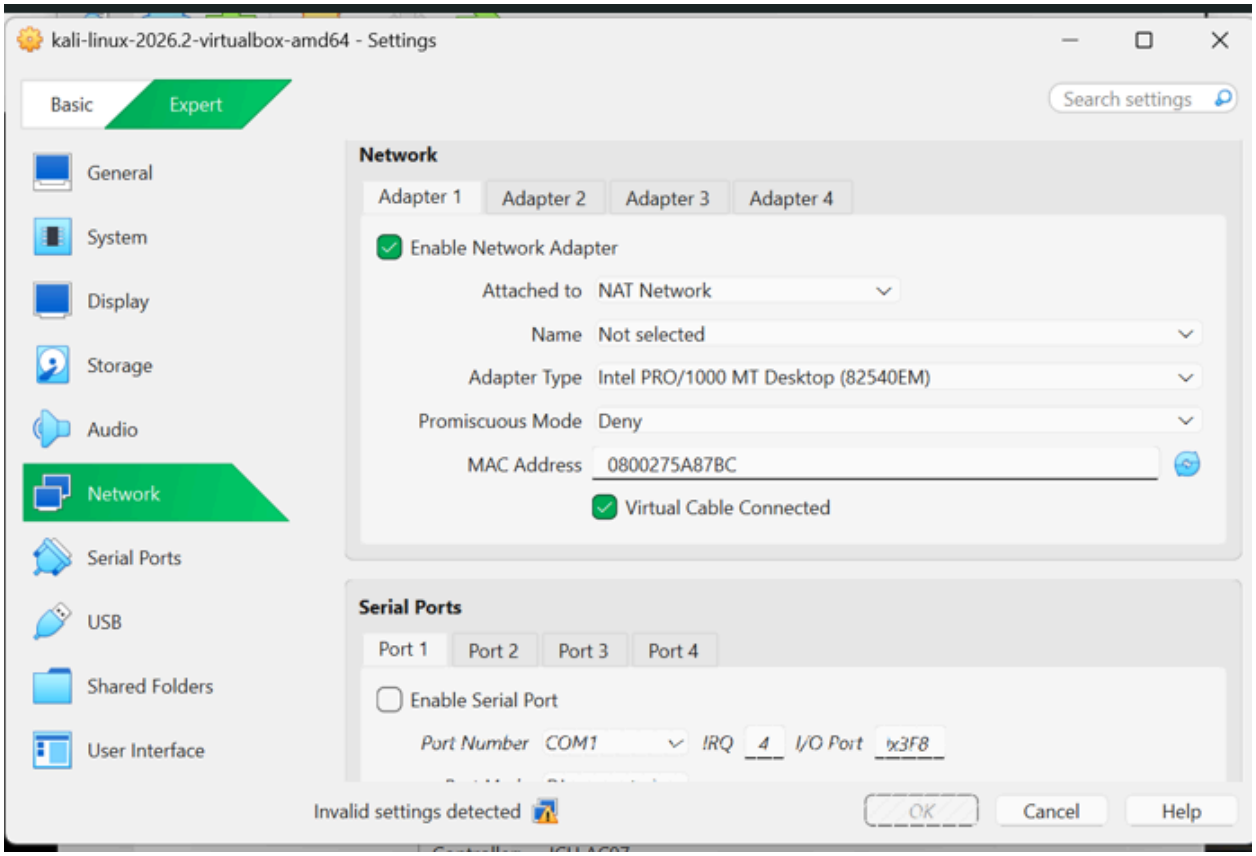
I set up the 1 st adapter to NAT mode.
Why NAT?

From **Oracle® VM VirtualBox User Manual for Release 6.0**

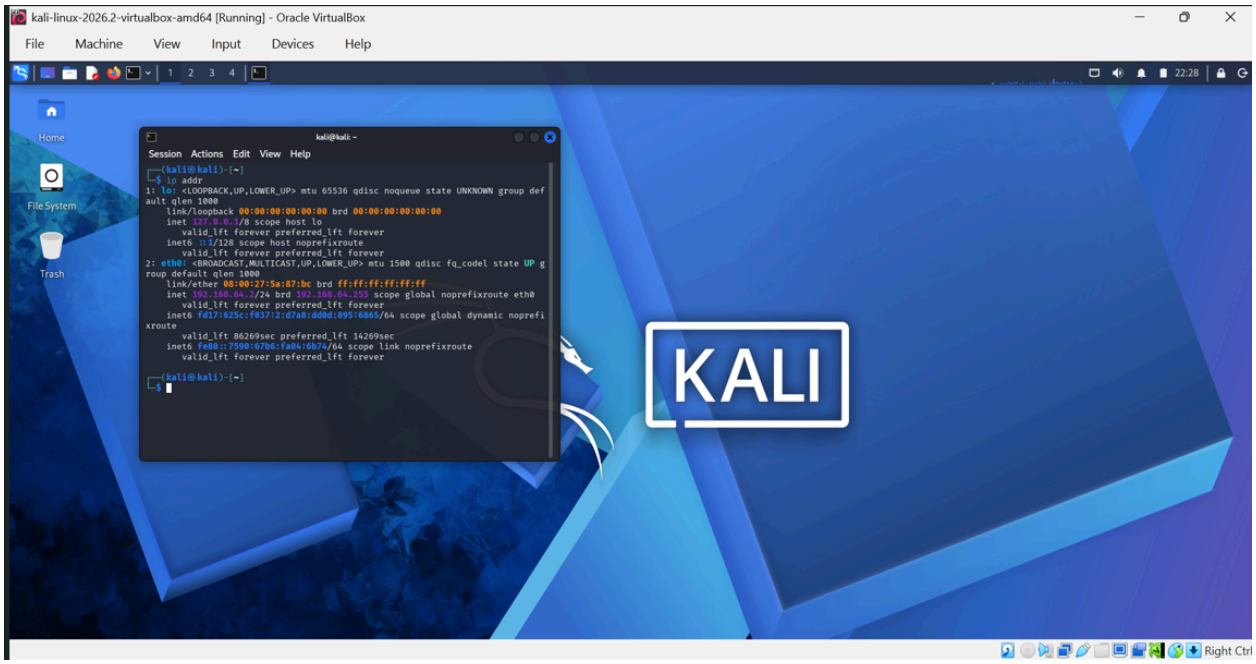
https://docs.oracle.com/en/virtualization/virtualbox/6.0/user/network_nat.html

Network Address Translation (NAT) is the simplest way of accessing an external network from a virtual machine. Usually, it does not require any configuration on the host network and guest system. For this reason, it is the default networking mode in Oracle VM VirtualBox.

A virtual machine with NAT enabled acts much like a real computer that connects to the Internet through a router. The router, in this case, is the Oracle VM VirtualBox networking engine, which maps traffic from and to the virtual machine transparently. In Oracle VM VirtualBox this router is placed between each virtual machine and the host. This separation maximizes security since by default virtual machines cannot talk to each other.



If we reboot Kali Virtual machine we can try to check the ip address by typing in Kali's terminal the command:
Ip addr



Nmap and Discovery

1. Ping scan / host discovery — what you just ran

nmap -sn <target/range>
Nmap -sn 192.168.78.5

No port scanning at all — just finds what's alive. This is almost always your first move on a new network. On a local subnet like yours, Nmap uses ARP requests under the hood (faster and more reliable than ICMP on a local segment).

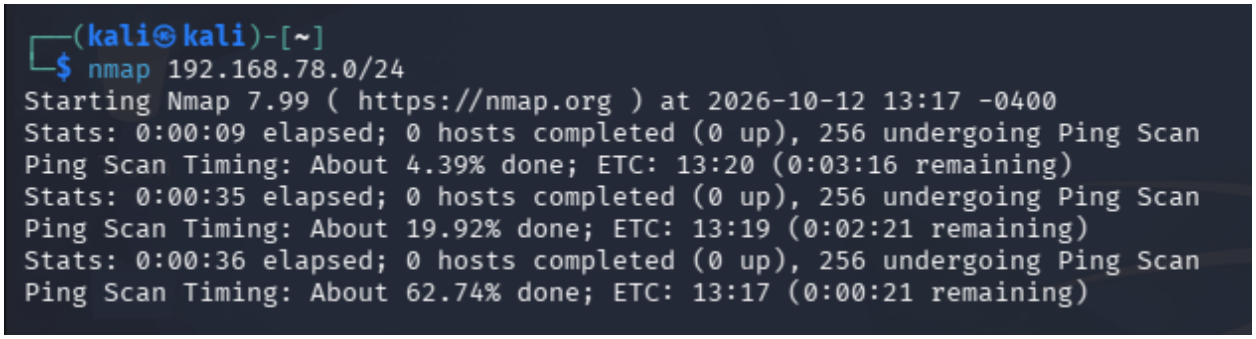
```
┌──(kali㉿kali)-[~]
└─$ nmap -Pn 192.168.78.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-12 13:46 -0400
Nmap scan report for 192.168.78.5
Host is up.
All 1000 scanned ports on 192.168.78.5 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 203.62 seconds
```

2. Default scan (top 1000 ports + service guess)

nmap <target>

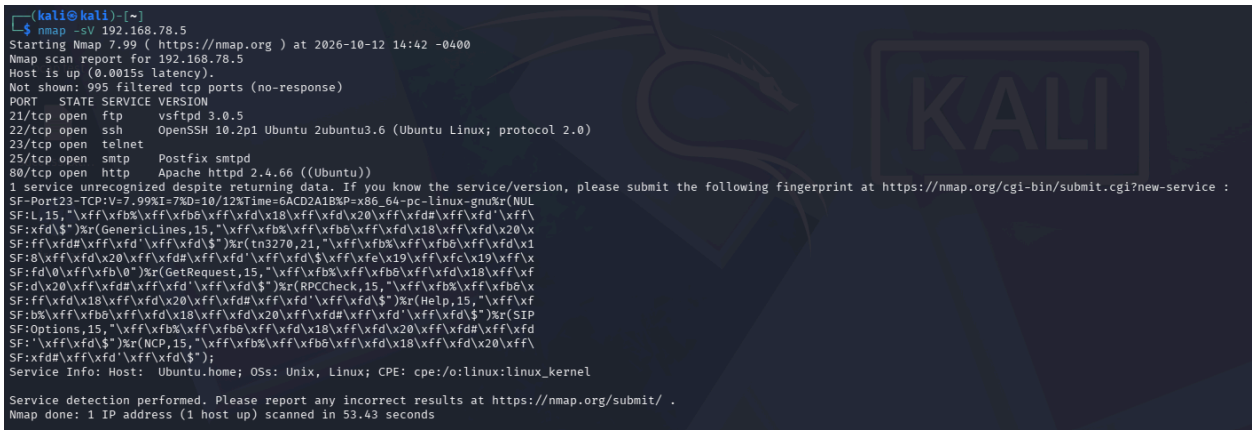
Without any flags, Nmap scans the 1000 most common TCP ports and reports open/closed/filtered. This is the “just tell me what’s open” command most people reach for once they know a host exists.



3. Service/version detection

nmap -sV <target>

Goes further than just “open” — tries to identify what software and version is actually listening on each open port (e.g., “vsftpd 2.3.4” rather than just “port 21 open”). This is the command that actually gives you something to look up in a vulnerability database.



4. Default scripts + version detection (very commonly paired together)

nmap -sV -sC <target>

`-sC` runs Nmap's built-in script library's "safe" default scripts — things like grabbing banners, checking for common misconfigurations, pulling extra metadata. Paired with `-sV`, this is probably the single most common "give me a solid first look at this host" command in practice.

5. OS detection

sudo nmap -O <target>

Guesses the operating system based on how the target's network stack responds to specific crafted packets. Needs root/sudo because it requires raw packet access.

6. Full port range

sudo nmap -p- <target>

Scans all 65535 ports instead of just the top 1000. Slower, but catches anything running on an unusual port that the default scan would miss.

7. The "give me everything" combo — often used once you've already confirmed a host is worth a deep look:

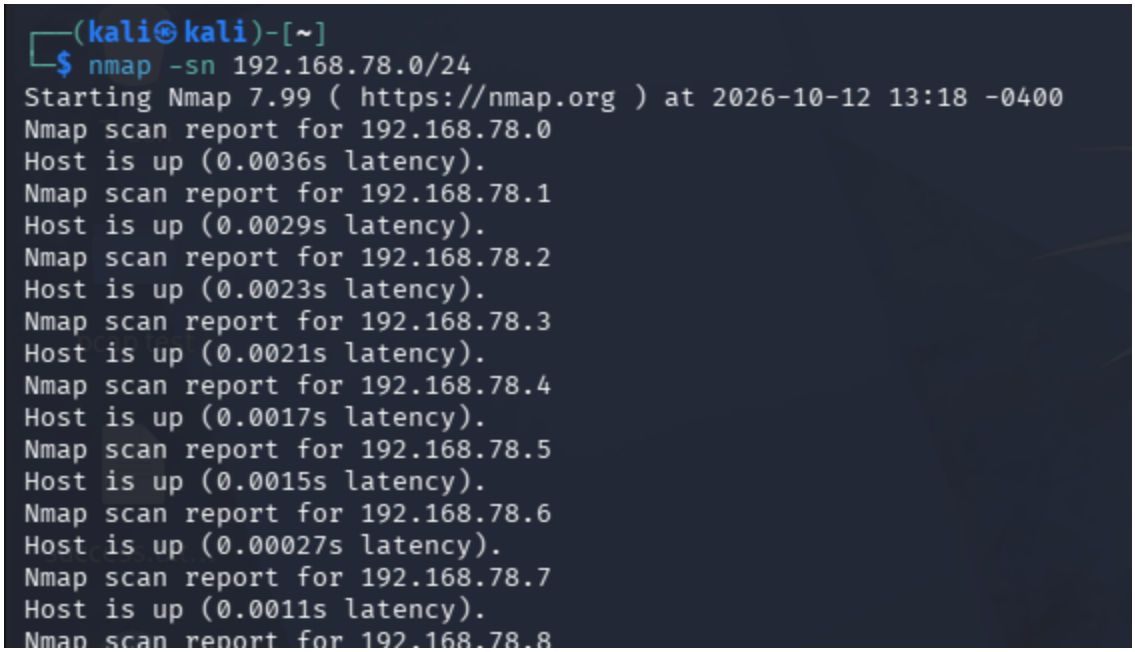
sudo nmap -sV -sC -O -p- <target>

8. UDP scan (less common, but worth knowing)

sudo nmap -sU <target>

TCP gets all the attention, but some services (DNS, SNMP, DHCP) run over UDP — this catches those. Slower and often skipped unless the rubric specifically wants broader coverage.

For your reconnaissance report, the natural sequence is: `-sn` (discovery, what you just did) → `-sV -sC` (identify services) → `-O` (fingerprint OS) → `-p-` (confirm nothing's hiding on an odd port). That's exactly the four-step enumeration structure already in your Target Profile template.



What `-sn` does. This is a “ping scan” (sometimes called host discovery) — it doesn’t check for open ports at all. It just sweeps the whole subnet asking “is anything here?” using ARP requests (on a local Ethernet segment like yours) rather than ICMP. The output tells you which IP addresses have something live behind them, nothing more — no services, no OS, no vulnerabilities yet. This is step one of reconnaissance: find out what’s actually on the network before you probe anything in detail.